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Effect of Al Content on Inclusions in Fe-15Mn-*x*Al-C Low-Density Steel

Lei Cao, Guocheng Wang, Yuanyou Xiao, and Ke Tao*

Three experiments of Fe-15Mn-*x*Al-C ($x = 1, 3, 5$) low-density steel with different Al contents are carried out, and the characteristics of inclusions are observed through field-emission scanning electron microscopy. The results show that the types of inclusions in the three samples are almost the same, mainly Al_2O_3, AlN, MnS, TiN, Al_2O_3-MnS, AlN-MnS, TiN-MnS, and Al_2O_3-AlN-MnS. The number percentage of Al_2O_3 decreases, while the number percentage of AlN increases with the increase of Al content in the three samples. As the Al content increases, the number of inclusions between 1 and 5 μm decreases, while the number between 5 and 10 μm increases. FactSage8.2 software calculation shows that an increase in Al content leads to an increase in the liquidus and a decrease in the solidus of the steel; at the same time, it causes the austenite region to narrow and expands the ferrite region. The change in the solidification route leads to a delay in the solidification fraction of AlN, MnS, and TiN inclusions precipitation. At last, the 2D mismatch theory shows that the mismatch degrees of Al_2O_3/MnS, AlN/MnS, TiN/MnS, and AlN/Al_2O_3 are 8.41%, 8.94%, 11.65%, and 11.66%, respectively.

1. Introduction

In recent years, Fe–Mn–Al–C low-density steel has been applied in important fields such as automobiles, ships, and aerospace, due to its excellent wear resistance, corrosion resistance, and lightweight.[1–5] Mn element has the effect of expanding the austenite zone, and its induced twinning effect is beneficial for hindering the propagation of cracks in the material, significantly improving the plasticity of the steel.[6] When the Al content in the steel increases by 1%, the density of the steel decreases by about 1.3%.[7,8] The high content of Al can reduce the strain hardening rate of the steel, and improve the formability and low-temperature toughness.[9] However, in the smelting process of Fe–Mn–Al–C low-density steel, the high content of Mn and Al can have a serious impact on the precipitation of inclusions.

Currently, some work has been conducted on the impact of changes in Al content on the characteristics of inclusions in medium/high Mn steel. Yu et al.[10] found that the increase in Al content resulted in an increase in the types of inclusions, and the morphology evolution order of inclusions is: spherical → irregular (polyhedral) → spherical. Alba et al.[11] found that the content of Al increased from 0.5% to 3%, and the number percentage of AlN-MnS inclusions in Fe-5Mn-*x*Al-C steel sharply increased. Wang et al.[12] studied that the inclusion with the highest number of inclusions in steel changed from MnS to AlN, as the Al content increased from 0.87% to 6.76%. In addition, they found that AlN inclusions could form in the liquid phase when the Al content is higher than 3.28%. Xin et al.[13] believed that the number density of inclusions in Fe-16Mn-*x*Al-0.6C steel increased first and then decreased, while the size of inclusions showed a change of first decreasing and then increasing when the Al content increased from 0.002% to 2.1%. Park et al.[14] investigated that the number percentage of AlN inclusions in Fe-20Mn-*x*Al steels gradually increased as the Al content increased from 1% to 6%. Wang et al.[15] investigated that when the Al content increased from 0.002% to 2.1%, the main type of inclusions in Fe-16Mn-*x*Al-C steel changed from MnO/(MnO + MnS) to AlN. They also found that AlN inclusions can worsen the hot ductility of steel in samples with Al content of 0.033% and 0.54%. Wang et al.[16] studied that the main composite inclusions transform from Al_2O_3-MnS to AlN-MnS inclusions with the increase of Al content. At the same time, the number percentage of AlN-MnS inclusions first increases and then decreases. Meanwhile, Jin et al.[17] confirmed through experiments and theoretical calculations that AlN can provide heterogeneous nucleation sites for MnS inclusions. In our previous study, we reported that an increase in Mn content is beneficial for expanding the FCC zone and leading to an advance in the solidification fraction of AlN, MnS, and TiN inclusions precipitation.[18] However, there are few reports on the effects of changes in Al content on the solidification route and inclusion formation of Fe-15Mn-*x*Al-C steel.

In the current work, three smelting experiments of Fe-15Mn-*x*Al-C ($x = 1, 3, 5$) low-density steel with different Al contents were conducted under laboratory conditions to study the effect of Al content on the solidification route and inclusions formation

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of steel. The formation mechanism of various types of inclusions was discussed based on changes in Al content.

2. Experimental Section

2.1. Smelting Process

Three smelting experiments of Fe-15Mn-*x*Al-C ($x = 1, 3, 5$) low-density steel with different Al contents were conducted under laboratory conditions. The first smelting experiment process is as follows: First, iron powder (700 g) and ferrous oxalate ($FeC_2O_4 \cdot H_2O$, 1.575 g) are placed in an alumina crucible and then placed in a tubular furnace for heating. When the tubular furnace was heated to 1873 K, kept for 40 min, and then deoxidizer was added. First, 225 g of ferromanganese was added and kept for 20 min. Then 11 g of aluminum particles were added that were wrapped in the pure iron sheet and kept for 60 min. Finally, the sample was cooled to room temperature in a tubular furnace. The operating steps and the amount of ferromanganese added in the three groups of experiments were consistent. The mass of pure aluminum particles added to the second and third groups was 28 and 48 g, respectively. The chemical composition of iron powder, ferromanganese, and aluminum particles is shown in **Table 1**.

2.2. Analytical Methods

The chemical composition of three steels was detected using chemical analysis methods. The content of T.O and T.N was measured by a carbon sulfur analyzer. The carbon sulfur analyzer detected the content of C and S. Inductively coupled plasma mass spectrometry measured the content of Al, Si, Ti, Mn, and P.

The inclusions in the experimental samples were observed in two methods. 1) Metallographic observation. Solid steel was cut into cylinders using an electric discharge wire-cutting machine. Subsequently, metallographic sandpaper and metallographic polishing paste were used to process the sample into a mirror surface. 2) Preparation of inclusion powder by nonaqueous solvent electrolytic separation. After electrolytic separation, a magnet was used to separate the inclusion particles from the solution containing ethanol and collect them on a carbon strip.[19–21] The samples prepared by the above two methods were used for field-emission scanning electron microscopy (Model: ZEISS ΣIGMA HD) with energy dispersive spectrometer (Model: Oxford-X-Max 50 mm^2) of the morphology and type of inclusions. The size, type, and number of inclusions were detected using an automatic inclusion analysis system, with a detection area of 52 mm^2.

Table 1. Chemical content of experimental raw materials (mass percentage, %).

Raw material	C	Si	Mn	P	S	Ti	Al
Iron powder	0.35	0.01	0.046	0.022	0.013	0.0086	–
Ferromanganese	0.76	1.46	65.860	–	0.032	–	–
Aluminum particles	–[a)]	–	–	–	–	–	99.999

[a)]– Represents trace amounts.

3. Results

3.1. Chemical Composition

The chemical contents of the three steels are listed in **Table 2**. Three experimental steels were named based on their Al content: 1 # (Al = 0.868%), 2 # (Al = 2.766%), and 3 # (Al = 4.695%). The N content in steel decreases with the increase of Al content, while the O content slightly increases with the increase of Al content. The content of Si and Ti is ≈0.32% and 60 ppm.

3.2. Morphology and Type of Inclusions

Figure 1 shows the morphology and type of inclusions in 1 # sample. Figure 1a shows an elongated Al_2O_3 inclusion with a length higher than 8 μm. Figure 1b shows an approximately elliptical AlN inclusion. Figure 1c shows the extracted elongated AlN inclusion with a size of ≈12 μm. Figure 1d shows a single-particle MnS inclusion, and Figure 1e shows an angular TiN inclusion. Figure 1f shows Al_2O_3-MnS composite inclusion in the inner and outer layers. Figure 1g shows the element surface distribution of AlN-MnS inclusions. Figure 1h indicates the TiN-MnS composite inclusion, and it can be observed that there is a clear interfacial relationship between MnS and TiN inclusions. Figure 1i shows the 3D morphology of Al_2O_3-AlN-MnS inclusions. By scanning the elemental surface distribution in Figure 1i, the composition of the composite inclusions can be clearly observed.

Figure 2 shows the morphology and type of inclusions in 2 # sample. Figure 2a shows a single AlN inclusion with a size of ≈23 μm. Figure 2b shows the 3D of MnS inclusions with an octahedron. Figure 2c shows a single-particle TiN inclusion like a boot. Figure 2d shows the Al_2O_3-MnS composite inclusion, where black represents the Al_2O_3 inclusion and gray represents the MnS inclusion. Figure 2e shows the AlN-MnS inclusion, and it can be observed that there is a close binding relationship between MnS and AlN inclusions. Figure 2f shows the Al_2O_3-AlN-MnS composite inclusions, which are Al_2O_3, AlN, and MnS inclusions from the inside out. Figure 2g shows the TiN-MnS inclusion.

Figure 3 shows the 3D morphology and type of inclusions extracted from 3 # sample. Figure 3a,b shows AlN inclusions with single and multiple particles with polyhedral structures. Figure 3c,d shows octahedral MnS inclusions with a size of ≈10 μm. Figure 3e,f shows TiN inclusions, which exhibit elongated shapes with lengths higher than 30 μm. Figure 3g shows the Al_2O_3-MnS inclusion, with the Al_2O_3 inclusion located at the bottom and the MnS inclusion located above the Al_2O_3 inclusion.

Table 2. Chemical content of experimental steel (mass percentage, %).

No.	C	Si	Mn	Al	N	P	S	O	Ti	Fe
1 #	0.41	0.32	14.85	0.868	0.0032	0.043	0.014	0.0005	0.0057	Bal.
2 #	0.41	0.32	14.86	2.766	0.0029	0.045	0.014	0.0005	0.0059	Bal.
3 #	0.41	0.31	14.83	4.695	0.0021	0.045	0.016	0.0013	0.0060	Bal.

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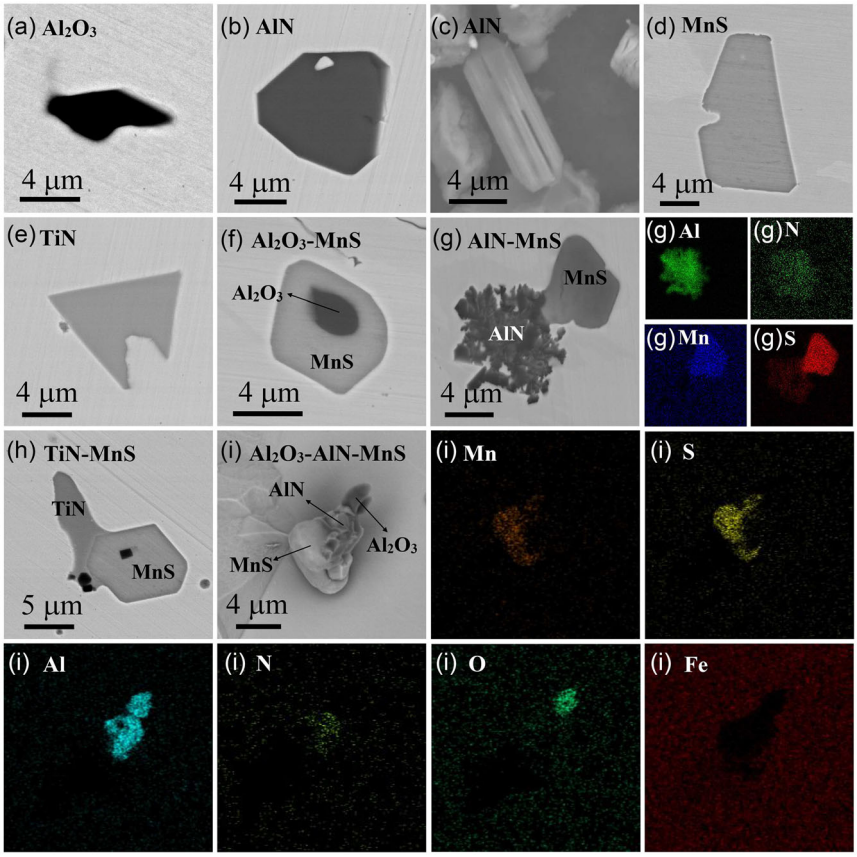


Figure 1. Morphology and type of inclusions in 1 # sample. a–e) single particle inclusions and f–i) composite inclusions.

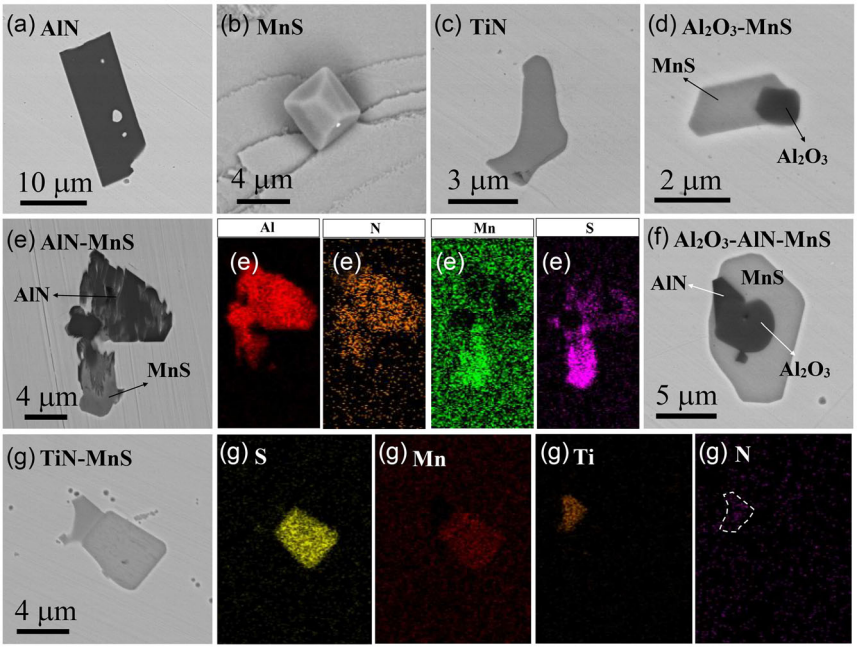


Figure 2. Morphology and type of inclusions in 2 # sample. a–c) single particle inclusions and d–g) composite inclusions.

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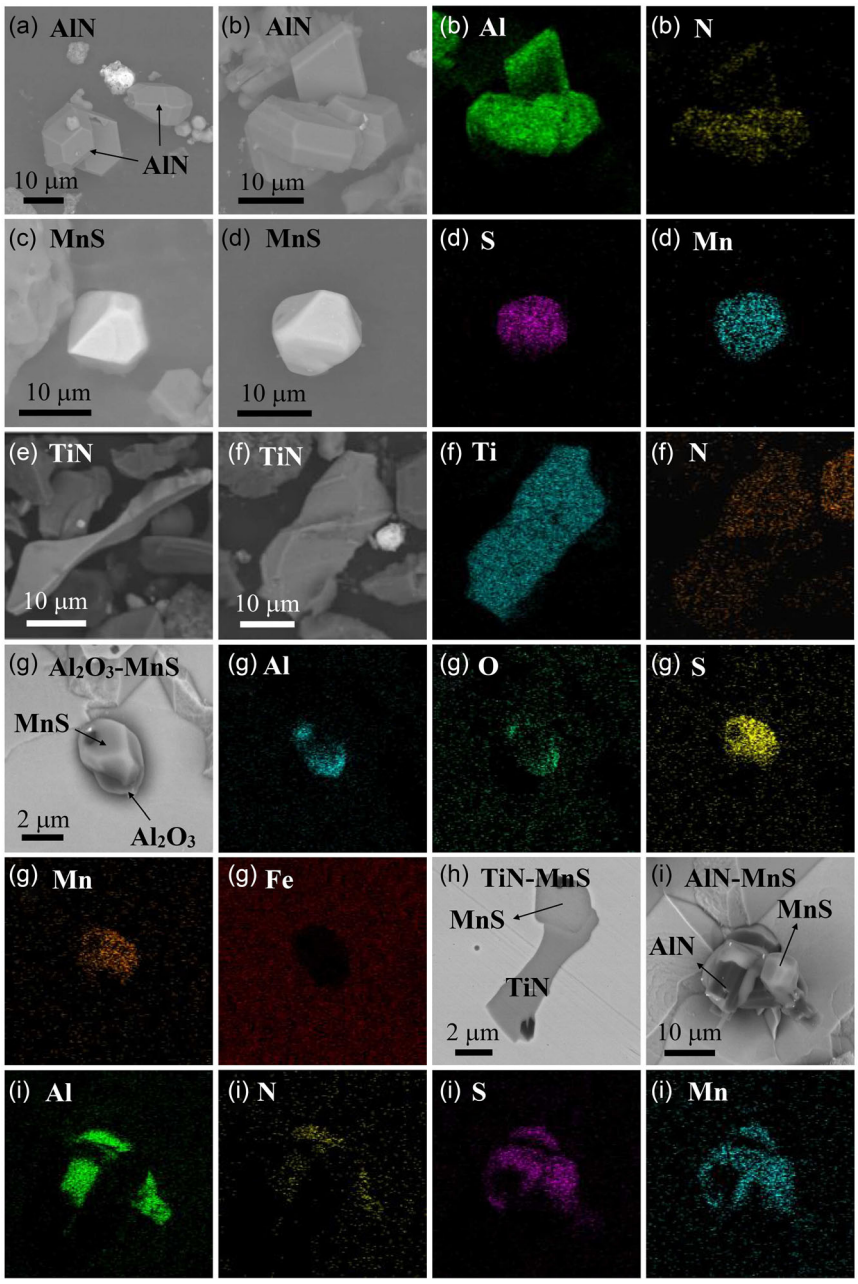


Figure 3. Morphology and type of inclusions in 3 # sample. a,b) AlN; c,d) MnS; e,f) TiN; and g–i) composite inclusions.

Figure 3h shows the TiN-MnS composite inclusion, whose morphology is the shape of a cigarette bag. Figure 3i shows the inclusion of AlN-MnS, and it is clear from morphology that the binding between MnS and AlN inclusions is very tight, and the connection between them is not like a physical way.

3.3. Number and Size Distribution of Inclusions

Figure 4 shows the number percentage of inclusions in 1 # to 3 # samples. From Figure 4, the number percentage of single Al_2O_3 inclusions in the steel decreases, the number percentage of single AlN inclusions gradually increases, while the number percentage of single MnS inclusions first increases and then decreases as the increasing of Al content. The number percentage of single TiN inclusions does not change significantly with the increase of Al content. The number percentage of Al_2O_3-MnS inclusions increases with the increase of Al, while the number percentage of AlN-MnS and TiN-MnS inclusions shows a decreasing trend. Al_2O_3-AlN-MnS inclusions exist in 1 # and 2 # samples but were not observed in 3 # sample.

Figure 5 shows the size distribution of inclusions in 1 # to 3 # samples. From Figure 5, it can be observed that the number of inclusions with sizes between 1 and 5 μm accounts for 58.4%, 53.6%, and 48.2% of the total number in 1 # to 3 # samples,

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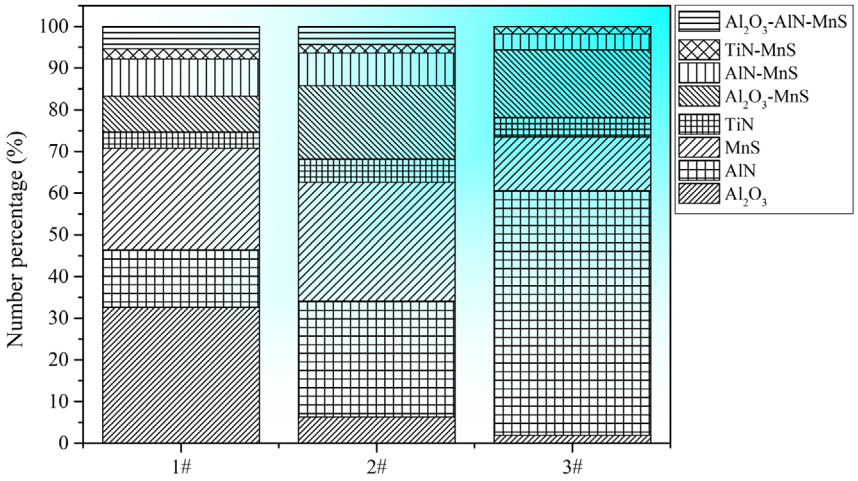


Figure 4. Number percentage of inclusions in three samples.

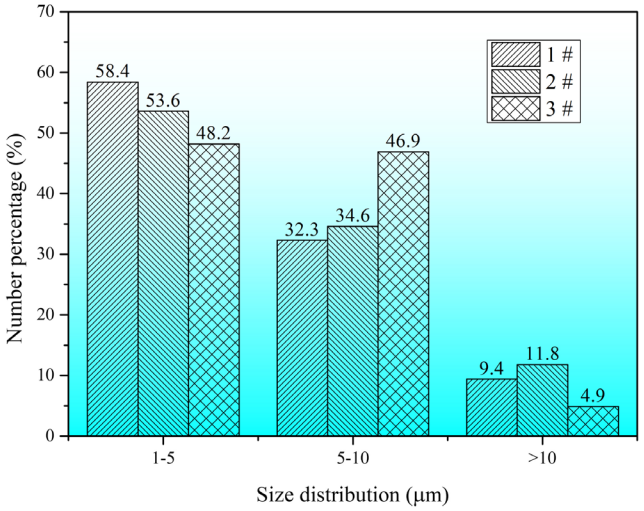


Figure 5. Size distribution of inclusions in three samples.

respectively. The number of inclusions with sizes between 5 and 10 μm accounts for 32.3%, 34.6%, and 46.9% of the total number in 1 # to 3 # samples, respectively. The number of inclusions with sizes higher than 10 μm accounts for 9.4%, 11.8%, and 4.9% of the total number in 1 # to 3 # samples, respectively. In 1 # to 3 # samples, the number of inclusions with sizes ranging from 1 to 5 μm decreases with increasing Al content, the number of inclusions with sizes ranging from 5 to 10 μm increases with increasing Al content, and the number of inclusions with sizes higher than 10 μm shows a trend of first increasing and then decreasing with increasing Al content.

4. Discussion

4.1. Effect of Al Content on the Solidification Route of Steel

To investigate the effect of Al content on the solidification route of three samples, the phase diagram module in FactSage 8.2 was used to calculate the Fe–C–Si–Mn–P–S–Al–O–N–Ti system and the calculation results are shown in **Figure 6**. **Table 3** shows the phase composition of the corresponding numbers in Figure 6. It can be seen that the solidification route of 1 # sample is: L → L + FCC → FCC, while the solidification route of 2 # and 3 # samples is: L → L + BCC → L + BCC + FCC → L + FCC → FCC and L → L + BCC → L + BCC + FCC → BCC + FCC → FCC, respectively. The increasing content of Al leads to the solidification route of 2 # and 3 # samples passing through the BCC phase, and the area where the FCC phase exists decreases.

The liquidus temperature and solidus temperature of the samples were obtained using the FactSage 8.2. The liquidus temperature (T_L) of 1 # to 3 # samples are 1702, 1740, and 1796 K, and the solidus temperature (T_S) are 1588, 1560, and 1532 K, respectively. The liquidus temperature increases with the increasing content of Al, while the solidus temperature decreases. The precipitation of inclusions is closely related to the liquidus temperature and solidus temperature. Therefore, it is necessary to discuss the effect of changes in Al content on the formation of various inclusions in the steel.

4.2. Effect of Al Content on Inclusion Formation

4.2.1. Effect of Al Content on the Formation of AlN Inclusions

The chemical reaction equation for the formation of AlN inclusions is shown in Equation (1)[22]

$$\begin{aligned} \mathrm{AlN}_{(s)} &= [\mathrm{Al}] + [\mathrm{N}] \cdot \Delta G^{\theta}_{\mathrm{AlN}} = -RT \cdot \ln K^{E}_{\mathrm{AlN}} \\ &= 303500 - 134.6T \ \mathrm{J\ mol^{-1}} \end{aligned} \tag{1}$$

When $a_{\mathrm{AlN}} = 1$, the K^{E}_{AlN} could be expressed as

$$K^{E}_{\mathrm{AlN}} = \frac{a_{[\mathrm{Al}]} \cdot a_{[\mathrm{N}]}}{a_{\mathrm{AlN}}} = a_{[\mathrm{Al}]} \cdot a_{[\mathrm{N}]} = f_{[\mathrm{Al}]} \cdot f_{[\mathrm{N}]} \cdot [\mathrm{pct\ Al}] \cdot [\mathrm{pct\ N}] \tag{2}$$

In the above equation, a_{AlN}, $a_{[\mathrm{Al}]}$, and $a_{[\mathrm{N}]}$ denote the activity of AlN, Al, and N in the steel. $f_{[\mathrm{Al}]}$ and $f_{[\mathrm{N}]}$ denote the activity coefficients of Al and N, and [pct Al] and [pct N] are the mass fractions

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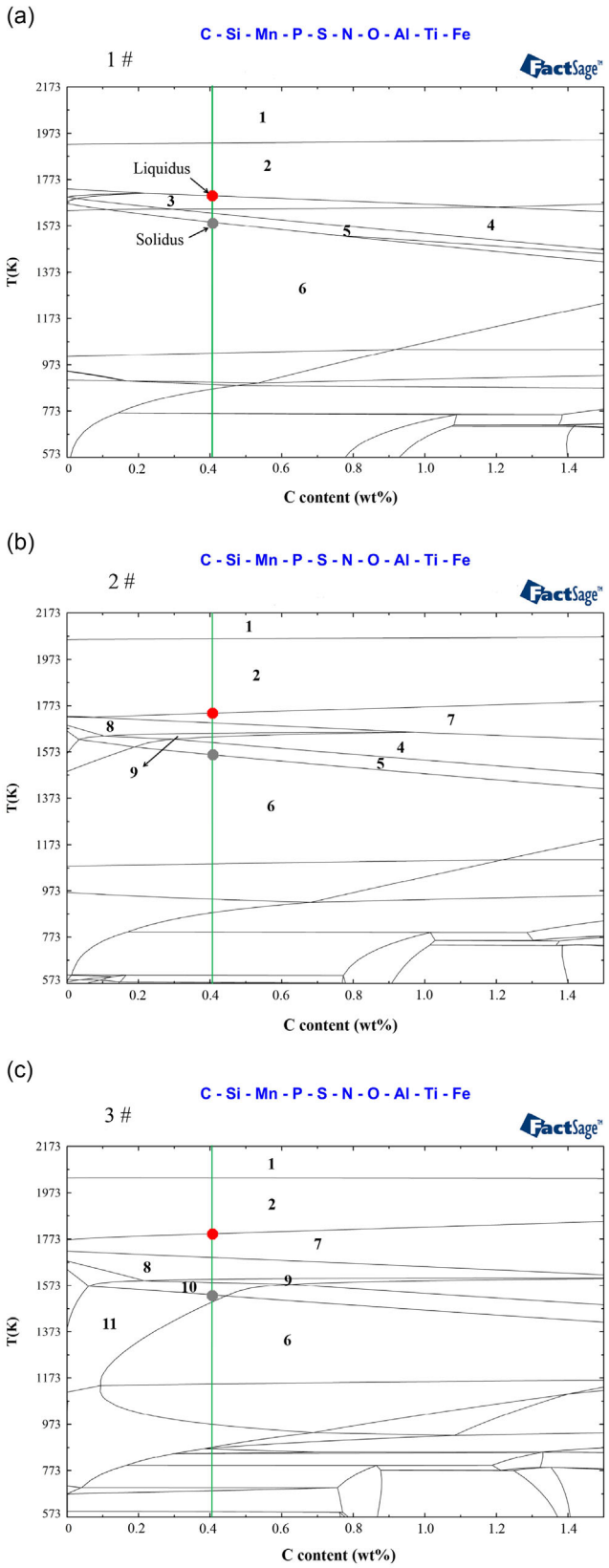


Figure 6. Calculation results of solidification routes, liquidus, and solidus temperatures for three samples. a) 1 #; b) 2 #; and c) 3 #.

of Al and N in the steel. Taking the logarithm at the base of 10 at both sites of Equation (2) and (3) can be obtained

$$\log K^{E}_{\mathrm{AlN}} = \log f_{[\mathrm{Al}]} + \log f_{[\mathrm{N}]} + \log([\mathrm{pctAl}] \cdot [\mathrm{pctN}]) \tag{3}$$

The calculation formula for the interaction coefficient is:

$$\log f_i = \sum e_i^j [\mathrm{pct}j] + \sum r_i^j [\mathrm{pct}j]^2 \tag{4}$$

where $i =$ Al, N, Ti, Mn, S; $j =$ C, Si, Mn, P, S, Al, N, O, Ti.

The interaction coefficients are shown in **Table 4**.[23–39] The theoretical concentration of AlN inclusions formed in 1 # to 3 # samples is expressed as:

$$\log([\mathrm{pctAl}] \cdot [\mathrm{pctN}]) = -\frac{15851}{T} + 7.225 \tag{5}$$

$$\log([\mathrm{pctAl}] \cdot [\mathrm{pctN}]) = -\frac{15851}{T} + 7.200 \tag{6}$$

$$\log([\mathrm{pctAl}] \cdot [\mathrm{pctN}]) = -\frac{15851}{T} + 7.187 \tag{7}$$

During the solidification stage, elements at the solidification front will be enriched due to segregation. Based on the Clyne–Kurz model,[40] the content of elements at the solidification front can be obtained:

$$C_{\mathrm{L}} = C_0[1 - (1 - 2\Omega k)z_{\mathrm{s}}]^{\frac{(k-1)}{(1-2\Omega k)}} \tag{8}$$

Among them, C_{L} represents the actual contents of elements at the solidification front, and C_0 represents the initial contents of elements in the steel. k represents the equilibrium distribution coefficient. z_{s} represents the solidification fraction. Ω denotes the inverse diffusion coefficient:[41]

$$\Omega = \alpha\left[1 - \exp\left(-\frac{1}{\alpha}\right)\right] - \frac{1}{2}\exp\left(-\frac{1}{2\alpha}\right) \tag{9}$$

$$\alpha = \frac{D \cdot t_{\mathrm{f}}}{(0.5\lambda)^2} \tag{10}$$

here D denotes the diffusion coefficient. The values of k and D are shown in **Table 5**.[42–44] t_{f} represents the solidification time. λ represents the distance between the secondary dendrite arms. The expressions for t_{f} and λ are as follows:[43]

$$t_{\mathrm{f}} = \frac{T_{\mathrm{L}} - T_{\mathrm{S}}}{R_{\mathrm{C}}} \tag{11}$$

$$\lambda = 143.9 R_{\mathrm{C}}^{-0.3616} \cdot [\mathrm{pct\,C}]^{(0.5501 - 1.996[\mathrm{pctC}])} \tag{12}$$

R_{C} represents the cooling rate.

The actual concentration product (Q_{AlN}) of AlN inclusion can be expressed as:

$$Q_{\mathrm{AlN}} = C_{\mathrm{L}}^{\mathrm{Al}} \cdot C_{\mathrm{L}}^{\mathrm{N}} \tag{13}$$

The relationship between solidification temperature (T_{f}) and solidification fraction is:

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Table 3. The phase composition corresponds to the number in Figure 6.

Number	Phase	Number	Phase
1	L	7	$L + Al_2O_3 + AlN$
2	$L + Al_2O_3$	8	$L + Al_2O_3 + AlN + BCC$
3	$L + Al_2O_3 + FCC$	9	$L + Al_2O_3 + AlN + BCC + FCC$
4	$L + Al_2O_3 + AlN + FCC$	10	$L + Al_2O_3 + AlN + BCC + FCC + MnS$
5	$L + Al_2O_3 + AlN + FCC + MnS$	11	$Al_2O_3 + AlN + BCC + FCC + MnS$
6	$Al_2O_3 + AlN + FCC + MnS$		

Table 4. First-Order and second-order interaction coefficient (1873 K).[23–39]

i	j								
	C	Si	Mn	P	S	Al	O	N	Ti
Al	0.091[24]	0.0056[24]	0[23]	–	0.03[24]	0.045[24]	−6.6[24]	−0.053[24]	–
	(−0.004)[25]	(−0.0006)[25]	(0)[23]	(−)[a)]	(−)	(−0.001)[25]	(39.823)[26]	(−0.004)[27]	(−0.0019)[28]
N	0.13[24]	0.047[24]	−0.021[24]	0.045[24]	0.007[24]	−0.028[24]	0.05[24]	0[24]	−0.53[24]
	(−)	(0.0013)[29]	(0)[30]	(−)	(−)	(0)[31]	(−)	(0)[31]	(0)[32]
Ti	−0.165[24]	0.05[24]	0.0043[24]	−0.0064[24]	−0.11[24]	–	−1.8[24]	−1.8[24]	0.013[24]
	(−)	(0)[33]	(−)	(−)	(−)	(0)[34]	(−0.355)[35]	(0)[36]	(0)[37]
Mn	−0.07[24]	–	–	−0.0035[24]	−0.048[24]	–	−0.083[24]	−0.091[24]	–
	(−)	(0)[38]	(0)[25]	(−)	(−)	(−)	(−)	(−)	(−)
S	0.11[24]	0.063[24]	−0.026[24]	0.029[24]	−0.028[24]	0.035[24]	−0.27[24]	0.01[24]	−0.072[24]
	(0.0058)[39]	(0.0017)[39]	(0)[39]	(0.0006)[39]	(−0.0009)[39]	(0.0009)[39]	(−)	(−)	(−)

[a)] (): r_i^j.

Table 5. The equilibrium distribution coefficient and diffusion coefficient.[42–44]

Element	$k(\delta)$	D^{δ}/[cm^2 s^{-1}]	$k(\gamma)$	D^{γ}/[cm^2 s^{-1}]
Al	0.6	5.9exp(−241417/RT)	0.6	5.9exp(−241417/RT)
N	0.25	0.008exp(−79078/RT)	0.48	0.91exp(−168490/RT)
Mn	0.77	0.76exp(−224430/RT)	0.785	0.055exp(−249366/RT)
S	0.05	4.56exp(−214639/RT)	0.035	2.4exp(−223426/RT)
Ti	0.38	3.15exp(−247693/RT)	0.33	0.15exp(−250956/RT)

$$T_f = T_0 - \frac{T_0 - T_L}{1 - z_s(T_L - T_S)/(T_0 - T_S)} \tag{14}$$

where $T_0 = 1809$ K.

The relationship among logK_{AlN}, logQ_{AlN}, and z_s can be obtained through Equation (5)–(14), as shown in **Figure 7**. In Figure 7, the solidification fractions of AlN inclusions precipitated in 1 # to 3 # samples are 0.46, 0.22, and 0.51, respectively. The solidification fraction of AlN inclusion precipitation advances from 0.46 to 0.22 with the increasing content of Al from 1% to 3%, and the solidification fraction of AlN inclusions precipitates is delayed from 0.22 to 0.51 with the increasing content of Al from 3% to 5%. The main reason for this phenomenon is the different degrees of segregation of Al and N during the solidification stages. From **Figure 8**, it can be seen that the degree of segregation of N element is much higher than that of Al element. Therefore, when there is no significant difference in N content in the steel, the increasing in content of Al promotes the precipitation of AlN inclusions. In 3 # sample, the solubility of N

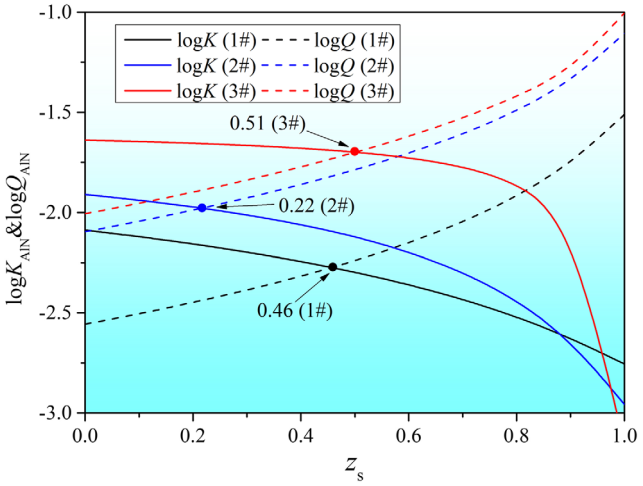


Figure 7. Relationship among logK_{AlN}, logQ_{AlN}, and z_s.

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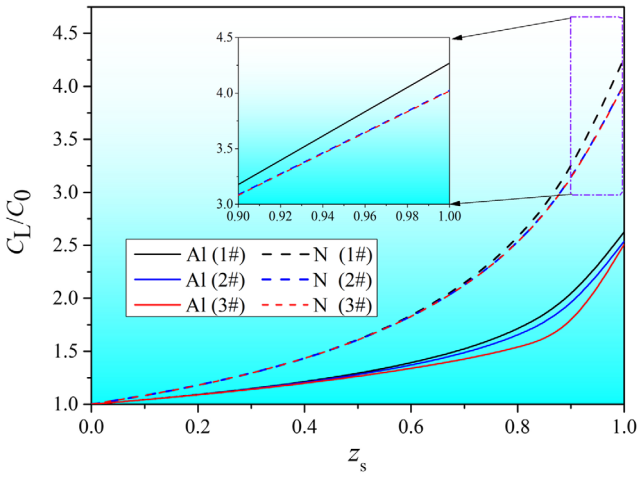


Figure 8. Al and N elements segregation ratios in three samples during the solidification stages.

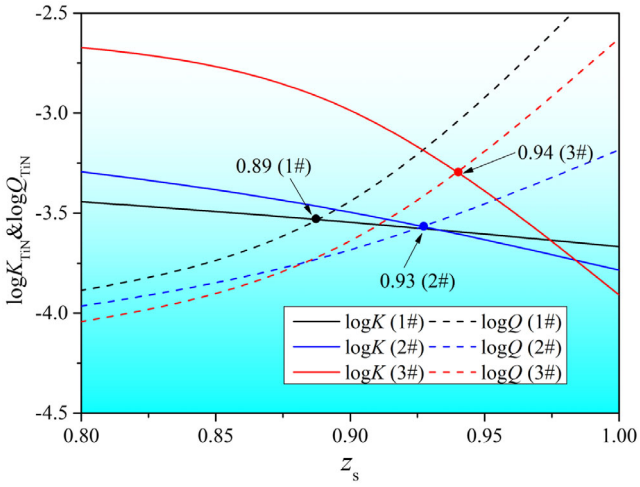


Figure 9. Relationship among $\log K_{TiN}$, $\log Q_{TiN}$, and z_s.

significantly decreased, leading to a delay in the solidification fraction of AlN inclusion precipitation. Consequently, the solidification fraction of AlN inclusion precipitation is delayed as a result of the combined action of the Al and N elements.

4.2.2. Effect of Al Content on the Formation of TiN Inclusions

The chemical reaction equation of TiN inclusions formation is expressed as follows[45]

$$\begin{aligned}\mathrm{TiN}_{(s)} &= [\mathrm{Ti}] + [\mathrm{N}] \cdot \Delta G^{\theta}_{\mathrm{TiN}} = -\mathrm{RT} \cdot \ln K^{E}_{\mathrm{TiN}} \\ &= 291000 - 107.91T\ \mathrm{J\,mol^{-1}}\end{aligned} \tag{15}$$

The relationship between the theoretical concentration of TiN inclusions formed in 1 # to 3 # samples and temperature (the thermodynamic calculation process is the same as the AlN inclusions):

$$\log([\mathrm{pctTi}] \cdot [\mathrm{pctN}]) = -\frac{15198}{T} + 5.902 \tag{16}$$

$$\log([\mathrm{pctTi}] \cdot [\mathrm{pctN}]) = -\frac{15198}{T} + 5.954 \tag{17}$$

$$\log([\mathrm{pctTi}] \cdot [\mathrm{pctN}]) = -\frac{15198}{T} + 6.009 \tag{18}$$

The relationship among $\log K_{TiN}$, $\log Q_{TiN}$, and z_s is shown in **Figure 9**. From Figure 9, the solidification fractions of TiN inclusions precipitated in 1 # to 3 # samples are 0.89, 0.93, and 0.94, respectively. Obviously, the increasing content of Al leads to a delay in the solidification fraction of TiN inclusions precipitation. The solidification fraction of TiN inclusion precipitation changes from 0.89 to 0.93 when the Al content increased from 1% to 3%, and the solidification fraction of TiN inclusion precipitation changes from 0.93 to 0.94 when the Al content increased from 3% to 5%.

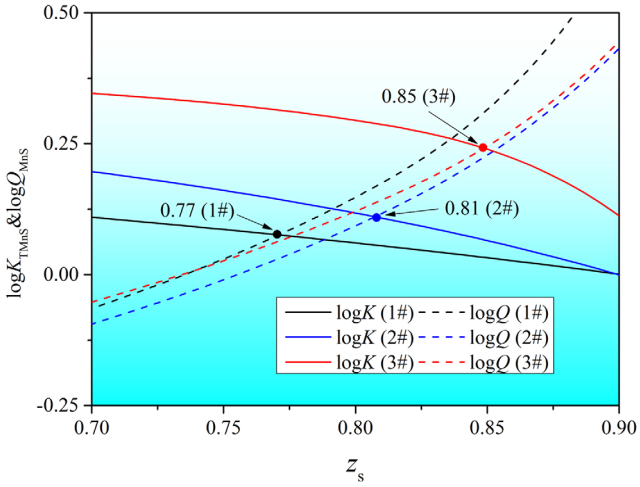


Figure 10. Relationship among $\log K_{MnS}$, $\log Q_{MnS}$, and z_s.

Table 6. Lattice parameters of different inclusion.

Inclusion	Lattice parameters [Å]	References
AlN	$a_{AlN} = b_{AlN} = 3.110$, $c_{AlN} = 4.980$	[48]
MnS	$a_{MnS} = b_{MnS} = c_{MnS} = 5.224$	[49]
Al_2O_3	$a_{Al_2O_3} = b_{Al_2O_3} = 4.759$, $c_{Al_2O_3} = 12.991$	[50]
TiN	$a_{MnS} = b_{MnS} = c_{MnS} = 4.242$	[5]

4.2.3. Effect of Al Content on the Formation of MnS Inclusions

The chemical reaction equation of MnS inclusions formation is shown in Equation (19):[46]

$$\begin{aligned}\mathrm{MnS}_{(s)} &= [\mathrm{Mn}] + [\mathrm{S}] \cdot \Delta G^{\theta} = -\mathrm{RT} \cdot \ln K^{E}_{\mathrm{MnS}} \\ &= 168822 - 98.87T\ \mathrm{J\,mol^{-1}}\end{aligned} \tag{19}$$

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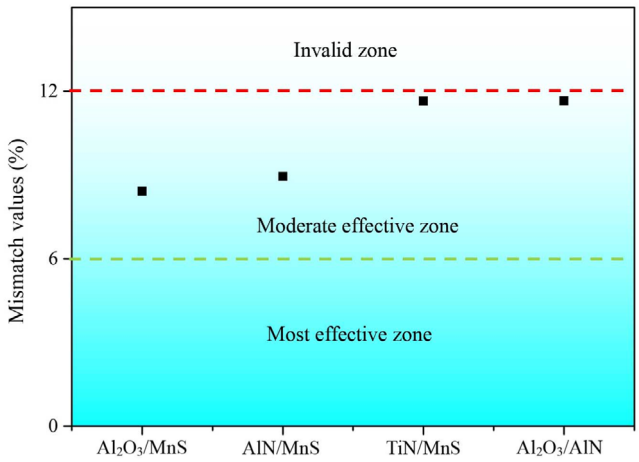


Figure 11. Calculation results of mismatch degree of composite inclusions.

The theoretical concentration product of MnS inclusions formed in 1 # to 3 # samples can be expressed by Equation (20)–(22):

$$\log([\text{pctMn}]\cdot[\text{pctS}]) = -\frac{8817}{T} + 5.482 \tag{20}$$

$$\log([\text{pctMn}]\cdot[\text{pctS}]) = -\frac{8817}{T} + 5.482 \tag{21}$$

$$\log([\text{pctMn}]\cdot[\text{pctS}]) = -\frac{8817}{T} + 5.331 \tag{22}$$

The relationship among $\log K_{\text{MnS}}$, $\log Q_{\text{MnS}}$, and z_s is shown in **Figure 10**. In Figure 10, the solidification fractions of MnS inclusions precipitated in samples 1 # to 3 # are 0.77, 0.81, and 0.85, respectively. It can be seen that the increasing content of Al leads to a delay in the solidification fraction of MnS inclusions precipitation. When the Al content increases from 1% to 5%, the solidification fraction of MnS inclusion precipitation changes from

Table 7. Calculation results of mismatch degree between different inclusions.

Interface	$MnS(100)/Al_2O_3(001)$			$MnS(110)/Al_2O_3(001)$			$MnS(111)/Al_2O_3(001)$		
$[uvw]_{MnS}$	[002]	[022]	[020]	[002]	[222]	[220]	[−101]	[−211]	[−110]
$[uvw]_{Al_2O_3}$	[−100]	[−110]	[010]	[−1−10]	[−200]	[−110]	[−100]	[−110]	[010]
d_{MnS} [Å]	5.224	7.388	5.224	5.224	9.048	7.388	3.694	6.398	3.694
$d_{Al_2O_3}$ [Å]	4.759	8.243	4.759	4.759	9.518	8.243	4.759	8.243	4.759
θ [°]	30	15	0	0	5.264	0	0	0	0
δ [%]		12.59			8.41			28.83	
Interface	$AlN(001)/Al_2O_3(001)$								
$[uvw]_{AlN}$		[−2−10]			[−300]			[−110]	
$[uvw]_{Al_2O_3}$		[−100]			[−110]			[010]	
d_{AlN} [Å]		5.387			9.330			5.387	
$d_{Al_2O_3}$ [Å]		4.759			8.243			4.759	
θ [°]		0			0			0	
δ [%]					11.66				
Interface	$MnS(100)/AlN(100)$			$MnS(110)/AlN(110)$			$MnS(111)/AlN(111)$		
$[uvw]_{MnS}$	[001]	[011]	[010]	[001]	[−111]	[−110]	[−101]	[−211]	[−110]
$[uvw]_{AlN}$	[011]	[020]	[01−1]	[001]	[−111]	[−110]	[−101]	[−211]	[−110]
d_{MnS} [Å]	4.069	5.754	4.069	4.069	7.048	5.754	2.877	4.984	2.877
d_{AlN} [Å]	3.705	5.240	3.705	5.240	9.076	7.410	3.705	6.418	3.705
θ [°]		0			0			0	
δ [%]		8.94			28.78			28.78	
Interface	(110)TiN//(100)MnS								
$[uvw]_{TiN}$		[001]			[−111]			[−110]	
$[uvw]_{MnS}$		[001]			[011]			[010]	
d_{TiN} [Å]		0.431			0.746			0.609	
d_{MnS} [Å]		0.537			0.759			0.537	
θ [°]		0			6.06			0	
δ [%]					11.65				

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0.77 to 0.85, and the delay interval is the same as the change interval of Al content. From the experimental results, it was found that the S content in 3 # sample was higher than that in 1 # and 2 # samples, while MnS inclusions were still delayed in precipitation.

4.3. Precipitation Mechanism of Complex Inclusion

Multiple types of composite inclusions were observed in the detection of experimental results, namely, Al_2O_3-MnS, AlN-MnS, Al_2O_3-AlN-MnS, and TiN-MnS. Through the observation results, it can be clearly observed that there is an interface relationship between the composite inclusions. The precipitation order of single-particle inclusions is: $Al_2O_3 > AlN > MnS > TiN$. Therefore, it can be speculated that the later formed inclusions can nucleate and grow at the heterogeneous nucleation sites of the first formed inclusions. To investigate the effectiveness of MnS inclusions nucleation on Al_2O_3, AlN, TiN surfaces, and AlN inclusions nucleation on Al_2O_3 surfaces, the mismatch degree (δ) as suggested by Bramfit,[47] substrates' capacity to promote heterogeneous phase nucleation was assessed. The calculation of mismatch is shown in Equation (23):

$$\delta_{(hkl)_n}^{(hkl)_s} = \frac{1}{3}\left[\sum_{i=1}^{3}\frac{|d^i_{[uvw]_s}\cos\theta - d^i_{[uvw]_n}|}{d^i_{[uvw]_n}}\right] \times 100\% \quad (23)$$

In the equation, the low-index lattice planes of the substrate and nucleation phase are represented by $(hkl)_s$ and $(hkl)_n$. The low-index crystal orientation in $(hkl)_s$ and $(hkl)_n$ is represented by $[uvw]_s$ and $[uvw]_n$, respectively. The interatomic spacing along $[uvw]_s$ and $[uvw]_n$ is represented by the $d^i_{[uvw]s}$ and $d^i_{[uvw]n}$, respectively. θ denotes the angle between $[uvw]_s$ and $[uvw]_n$.

The lattice parameters of different inclusions are shown in **Table 6**,[5,48–50] and the calculation results of mismatch degree at the interfaces of different inclusions are shown in **Figure 11**. From Figure 11, the mismatch degrees of Al_2O_3/MnS, AlN/MnS, TiN/MnS, and AlN/Al_2O_3 are 8.41%, 8.94%, 11.65%, and 11.66%, respectively. This indicates that Al_2O_3, AlN, and TiN could provide nucleation sites for the nucleation of MnS, and Al_2O_3 could also provide heterogeneous nucleation sites for AlN. The detailed calculation results are shown in **Table 7**.

Figure 12 shows the formation process of inclusions in three steels during smelting and solidification. A large number of single-particle Al_2O_3 inclusions are formed during the smelting process when the Al content is 1%. With the increasing content of Al, the precipitation amount of single-particle Al_2O_3 inclusions in the 3% and 5% samples decreases. In the subsequent solidification process, AlN, MnS, and TiN inclusions begin to precipitate, in the order of AlN > MnS > TiN. The solidification fraction of AlN inclusion precipitation is advanced with the Al content increased from 1% to 3%, and the solidification fraction of AlN inclusion precipitation is delayed with the Al content increased from 3% to 5%. The solidification fraction of MnS and TiN inclusions precipitated is delayed with the increasing content of Al. In solid steel, the first precipitated inclusions can provide nucleation sites for the later precipitated inclusions. Therefore, Al_2O_3-MnS, AlN-MnS, and TiN-MnS composite inclusions were observed in solid steel. Due to the presence of relatively small amounts of single-particle Al_2O_3 inclusions in samples containing 5% Al, no Al_2O_3-AlN-MnS inclusions were observed in the steel.

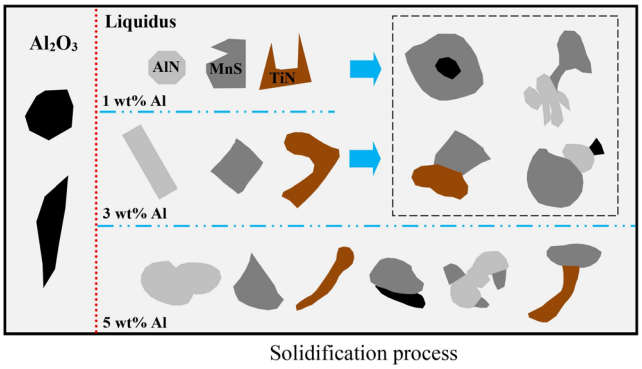


Figure 12. Evolution of inclusions during smelting and solidification process in three samples.

5. Conclusion

By studying the effect of different Al contents on the formation of inclusions in Fe-15Mn-*x*Al-C ($x = 1, 3, 5$) steel, the main conclusions are as follows: 1) The types of inclusions in Fe-15Mn-*x*Al-C ($x = 1, 3, 5$) low-density steel are almost the same, mainly Al_2O_3, AlN, MnS, TiN, Al_2O_3-MnS, AlN-MnS, TiN-MnS, and Al_2O_3-AlN-MnS. As the Al content in the three samples increases, the number percentage of Al_2O_3 decreases, while the number percentage of AlN increases. The number percentage of Al_2O_3-MnS inclusions increases with the increasing content of Al, while the number percentage of AlN-MnS and TiN-MnS inclusions decreases. As the Al content increases, the number of inclusions between 1 and 5 μm decreases, while the number between 5 and 10 μm increases; 2) The increasing content of Al leads to an increase in the liquidus and a decrease in the solidus of the steel. At the same time, it causes the FCC region to narrow and expands the BCC region, and the solidification route of steel samples with 3% and 5% Al content passes through the BCC region; 3) The changes in the liquidus, solidus, and solidification route of the sample result in changes in the precipitation time of inclusions. When the Al content increases from 1% to 3%, the solidification route changes from FCC to FCC and BCC, which leads to an advance in the solidification fraction of AlN precipitation. Furthermore, when the Al content increases from 3% to 5%, although the solidification route is still FCC and BCC, high content of Al leads to a significant decrease in N solubility. Therefore, the comprehensive results of Al and N elements delay the precipitation of AlN. However, the solidification fraction of MnS and TiN inclusions precipitated is delayed with the increasing content of Al; and 4) The mismatch degrees of Al_2O_3/MnS, AlN/MnS, TiN/MnS, and AlN/Al_2O_3 are 8.41%, 8.94%, 11.65%, and 11.66%, respectively. This indicates that Al_2O_3, AlN, and TiN could provide nucleation sites for the nucleation of MnS, and Al_2O_3 could also provide nucleation sites for AlN.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

Al contents, inclusions, low-density steels, thermodynamics

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